# Hub Cultural: Mapping Cultural Event Networks in France through Instagram Graph Mining and Network Analysis

From a Diaspora Network to a General Cultural Discovery Platform

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**Abstract**

**[TODO:** 150–250 words. Draft last, once Results/Discussion are stable. Should state: (1) the problem — culturally-relevant activity in the Paris region (with priority given to photography, cinema, literature, theatre, and local community events) is not systematically documented or discoverable, and this project originated from, but is no longer limited to, the specific case of the Latin American diaspora; (2) the method — a pipeline that scrapes Instagram, builds a Neo4j graph, applies network analysis (PageRank, betweenness, Leiden community detection) and an LLM/NLP event-extraction pipeline, combined with human curation as the trust signal, plus a research-operations discipline (decision log, run log, dry-run-first design) treated as methodology in its own right; (3) headline findings — network size and structure, the bridge role of commercial/institutional accounts, event landscape, and a transparent account of data-quality limits; (4) the practical artifact — a public, bilingual event-discovery website; (5) contribution — methodological (human curation vs. automated scoring; a cultural-relevance criterion replacing a nationality-based one) and empirical (first mapped network of this kind for this cultural ecosystem in France). **]**

# Contents

# Chapter 1

# Introduction

## 1.1 Context and Motivation

Hub Cultural began as an attempt to make visible something that is well known anecdotally but poorly documented empirically: the cultural life that Latin American, and in particular Colombian, communities sustain in the Paris region through small associations, independent venues, cultural attachés, and a dense but scattered presence on Instagram. The author's own Colombian background supplied both the initial motivation and the first seed account for data collection — the Colombian consulate in Paris (`@consuladocolparis`) — from which the network was grown outward through automatic relationship discovery.

As the project progressed, the boundary that had organized it from the start — *is this account or event part of the Latin American diaspora?* — proved both practically brittle and conceptually narrower than what the data, and the underlying research interest, actually supported. Many of the best-connected cultural accounts discovered through the network turned out to be French institutions, mixed-nationality collectives, or venues with no diaspora affiliation at all, but which were unmistakably part of the same cultural ecosystem the project set out to map — programming photography exhibitions, independent cinema, literary events, and theatre alongside explicitly Colombian or Latin American programming. The project's scope therefore evolved in two steps: first from Colombia-specific to Latin-American-diaspora-wide, and subsequently from a nationality-based criterion to a purely cultural one. The operative question became not *who is behind this account* but *is this culturally relevant*, with explicit priority — inherited from the project's original four thematic agendas (`docs/objetivos.md`) — given to photography, cinema, literature, and theatre, alongside a fifth, less formally bounded category: events organized by and for local communities, regardless of national or cultural origin.

This evolution is treated in this thesis not as scope creep to be apologized for, but as a methodological finding in its own right (§6.2): a nationality-based inclusion criterion turned out to be both harder to operationalize consistently and less faithful to how cultural networks actually cohere in practice, where diaspora and host-society cultural production overlap and reinforce each other rather than remaining cleanly separable.

## 1.2 Problem Statement and Research Question

The cultural activity produced by, and adjacent to, migrant and diaspora communities in a host country is rarely centralized: it lives across dozens of Instagram accounts, is announced inconsistently, and is legible mainly to whoever already follows the right accounts. This is true of Latin American cultural life in the Paris region specifically, but — as this project's own data eventually made clear — it is not a problem unique to any one national community; it is a structural feature of how small-scale cultural production circulates on social media generally.

This thesis asks: *How is culturally-relevant activity — with priority given to photography, cinema, literature, theatre, and local community cultural life — organized as a network of accounts and events in the Paris region, and can that structure be systematically mapped, analyzed, and made discoverable using publicly available Instagram data?*

This breaks into three sub-questions:

1. **Structural.** Who are the connective actors in this network — which accounts function as hubs or bridges between otherwise separate cultural circles, and does this differ between commercial/institutional accounts and individual or grassroots ones?

2. **Thematic.** How is cultural activity distributed across the project's priority domains (photography, cinema, literature, theatre, community events), and how well does an automated pipeline classify it into these domains compared to human judgment?

3. **Practical.** Can the resulting graph be turned into a public-facing tool that helps the community it describes actually discover and attend these events — and what does building that tool reveal about the reliability of the underlying data?

## 1.3 Objectives

The project's objectives, as originally framed (`docs/objetivos.md`) and updated here to reflect the current, cultural-relevance-based scope:

1. Build and maintain four thematic monitoring agendas — *literature*, *photography*, *cinema*, and *theatre* — tracking the organizations that play an active role in each, complemented by a fifth, less formally bounded priority: cultural events organized by and for local communities.

2. Consolidate a pipeline that extracts events from the accounts under these agendas, primarily from Instagram and, prospectively, from websites and other social networks.

3. Map the network of actors within each thematic agenda and categorize them — by type of organization, geographic zone, cultural identity where relevant, and structural role in the network.

4. Evaluate the participation opportunities these networks open up, using a deliberately human, non-technical success criterion: on this dimension, the project succeeds when it leads to actually meeting someone through it.

## 1.4 Scope and Contributions

**Scope.** Geographically the project centers on Île-de-France, though a meaningful share of curated accounts and events originate from, or are organized by, institutions physically located outside France — a scope boundary discussed explicitly in §3.11 rather than silently discarded. Instagram is the sole data source, a deliberate boundary rather than an oversight: it is the platform where this cultural activity is, in practice, announced. Only public accounts are in scope. Inclusion is governed by cultural relevance rather than national origin, with the four thematic pillars and the community-events category acting as explicit priorities rather than hard filters.

**Contributions.** This thesis makes three kinds of contribution:

- *Methodological* — a working pipeline combining network analysis, multilingual NLP event extraction, and human curation as the operative trust signal, arrived at through an evidenced pivot away from fully automated account scoring (§3.7); and a research-operations practice — a running decision log, a run log, and a dry-run-first discipline applied to every script — treated here as part of the methodology rather than incidental bookkeeping (§3.2).

- *Empirical* — a systematic mapping of this cultural network's structure: node and edge counts, community structure, and the role of bridging actors, alongside a transparent account of where the underlying data is, and is not, reliable (geocoding confidence, date-extraction accuracy, categorization accuracy).

- *Applied* — a public-facing, bilingual event-discovery website that turns the graph into something a member of the community it describes can actually use, not only an analytical artifact for the researcher.

## 1.5 Thesis Structure

Chapter 2 situates the project within diaspora and migration studies, social network analysis, computational social science, and NLP-based event extraction. Chapter 3 details the methodology, including the two methodological pivots this thesis treats as contributions in their own right: human curation as the trust signal, and a research-operations discipline of decision logging, run logging, and dry-run-first iteration. Chapter 4 describes the resulting system, from data collection through the public discovery website. Chapter 5 reports results: network structure, thematic and geographic distribution, and a transparent accounting of data quality. Chapter 6 discusses what these results mean, including a dedicated reflection on the project's evolution from a nationality-based to a cultural-relevance-based scope. Chapter 7 concludes, states limitations, and outlines future work.

# Chapter 2

# Literature Review / Theoretical Framework

## 2.1 Diaspora Studies and Cultural Networks

**[TODO:** Situate within diaspora/migration studies: how migrant communities organize culturally, the role of associations and informal networks, prior work on Latin American diasporas in Europe specifically if available. Framing note: this remains one relevant thread — the project's motivating case and a meaningful share of its curated content — but since the operative inclusion criterion is now cultural relevance rather than national origin (§6.2), pair this with literature on cultural-sector/creative-industry network mapping more generally, so the theoretical framing matches the project's actual scope and not only its origin story. **]**

## 2.2 Social Network Analysis in Migration Contexts

**[TODO:** Prior applications of SNA to migrant/diaspora networks; concepts to define here for later use: centrality (degree, betweenness, PageRank), community detection, structural holes / bridging, E-I index. **]**

## 2.3 Computational Social Science and Social Media as Data

**[TODO:** Justify Instagram as a data source; discuss known biases (who is/isn't represented on Instagram, algorithmic visibility, platform ToS and ethical use of public data). **]**

## 2.4 Graph Databases and Network Science Methods

**[TODO:** Neo4j / property graph model; why a graph database over a relational one for this problem; brief technical grounding for PageRank, betweenness centrality, Leiden algorithm, k-core decomposition, participation coefficient — cite original papers. **]**

## 2.5 NLP for Event Extraction and Classification

**[TODO:** Ground the 3-layer detection design: sentence embeddings for candidate filtering, NLI zero-shot classification, LLM-based extraction/classification/gating. Cite relevant NLP literature (semantic similarity, zero-shot NLI, LLM information extraction) and note the multilingual requirement (ES/FR/EN) as a specific design constraint. **]**

## 2.6 Positioning

**[TODO:** One paragraph: how this thesis differs from prior diaspora network studies (data-driven, automated + human-curated hybrid, produces a live public tool, not just an analysis) and from prior event-extraction NLP work (domain-specific gating logic, human-curation-as-trust-signal pivot). **]**

# Chapter 3

# Methodology

## 3.1 Research Design Overview

**[TODO:** High-level pipeline diagram description (7 phases): extraction → ingestion → network analysis → manual categorization → event enrichment → geo enrichment → cleanup. Figure: reproduce/adapt the pipeline diagram from `docs/data_flow.md`. **]**

## 3.2 Research Operations: Decision Logging, Run Tracking, and Iterative Safety

Independently of the technical pipeline itself, this project maintained three interlocking practices for managing the research process, treated here as a methodological contribution rather than incidental project management. Together they make the pipeline's evolution auditable after the fact — which is not free in a setting where the object of study is a continuously growing graph, reprocessed repeatedly by successive versions of the extraction logic.

**A real-time decision log.** Every non-trivial design choice was recorded, at the moment it was made, in a sequentially numbered decision log (`docs/decisions.md`, entries DD-001 through DD-044 and counting) rather than reconstructed retrospectively for this thesis. Each entry states the alternatives considered, the evidence that motivated the final choice, and, where applicable, validation results. Three entries illustrate the range of what this captured: DD-041 documents a geo-conflict guardrail added to the event-deduplication resolver after a production dry-run surfaced a specific failure mode — two structurally similar events merged despite being in different cities — validated against a 14-case test suite before being applied to 68 pending merges; DD-042 documents the addition of a per-event, LLM-generated tag field designed specifically to avoid a parsing bug present in an earlier, account-level free-text field; DD-043 and DD-044 document the design of the Neo4j-to-static-JSON export and the resulting public website, including an explicit accounting of which ranking components could be precomputed server-side versus which needed to remain client-side. Maintaining this log in real time, rather than writing it up after the fact, is itself a methodological choice: it captures the reasoning available *at the time* of each decision, including alternatives that were considered and rejected — something retrospective documentation systematically loses.

**A run log.** Separately, every substantive execution of the pipeline against live data — not every script invocation, but every run whose output was retained or acted upon — was logged with its date, parameters, and a summary of results (`docs/runs_log.md`). This served two purposes beyond record-keeping: it made it possible to distinguish a genuine change in the underlying data (new accounts or events discovered) from a change in pipeline logic when comparing two exports, and it created a paper trail for decisions that depended on a specific run's output rather than on memory — for instance, the decision not to add a venue-specificity guardrail to the event resolver was made with explicit reference to a logged run's results.

**Dry-run-first, diagnose-first design.** Every script that writes to the live Neo4j graph was built with a `-dry-run` or `-diagnose` mode as a first-class feature, not an afterthought: `cleanup_legacy_accounts.py` runs its deletion logic inside a single transaction and issues a `ROLLBACK` under `-dry-run`, so a preview's reported counts are exact rather than estimated; `load_manual_account_categorization.py` and the event-extraction and event-resolution scripts all follow the same pattern. Combined with the idempotent design of the extraction and ingestion scripts — re-running them is always safe, since already- processed accounts and posts are skipped or merged rather than duplicated — this made it possible to iterate on pipeline logic against a live, shared, non-trivially-sized research database without the usual risk of an irreversible mistake, a constraint that shaped several of the decisions recorded in the log above.

Together, these three practices form a coherent research-operations discipline: decisions are recorded when made, runs are recorded when executed, and changes to the live data are always previewable before being committed. For a thesis built on a continuously evolving dataset rather than a fixed one, this discipline is what makes it possible to state, for any given result in Chapter 5, exactly which version of the pipeline and which run produced it.

## 3.3 Data Collection

**[TODO:** Apify actors used (instagram-profile-scraper, instagram-post-scraper); seed selection strategy — originally `@consuladocolparis` as single seed, BFS-style expansion via `RELATED_TO` suggestions, later curated seed files (`config/seeds_*.json`); cost control mechanism (`.apify_cost_log.json`); incremental/idempotent design (`extract_profiles.py` only scrapes accounts not yet seen). **]**

## 3.4 Data Model

**[TODO:** Full Neo4j schema: node labels (`:Account`, `:Post`, `:Hashtag`, `:Location`, `:Track`, `:Comment`, `:IgtvVideo`, `:Event`, `:City`, `:Country`, `:Arrondissement`) and relationship types (`PUBLISHED`, `HAS_HASHTAG`, `MENTIONS`, `TAGS_USER`, `COAUTHORED_BY`, `MENTIONS_EVENT`, `ORGANIZED`, `PARTICIPATED_IN`, `LOCATED_IN`, etc.). Include an entity-relationship diagram (Appendix A). **]**

## 3.5 Network Analysis Methods

**[TODO:** Multiplex network design: social layer (MENTIONS/ TAGS_USER/COAUTHORED_BY, author→post→target projection) separate from the algorithmic layer (`RELATED_TO`, `Algo` suffix). Metrics computed: exact PageRank, exact betweenness, Leiden community detection at three resolutions ($\gamma = 0.5, 1.0, 1.5$, seed 42), weakly connected components, k-core, participation coefficient, E-I index by actor type. Note the pivot from Neo4j GDS (blocked — standard AuraDB lacks the plugin, corporate firewall also blocks port 7687) to a local igraph/leidenalg implementation (`3_analyze_network.py`) as a methodological adaptation worth documenting, not hiding. **]**

## 3.6 Event Detection Pipeline

**[TODO:** Describe the 3-layer architecture as it exists today (post the 2026-08 rewrite that removed the old embeddings-based multi-label typification layer after a 100-post human-vs-pipeline evaluation — cite `data_processed/eval_100_report.md`):

1. Layer 1 — cosine similarity against ∼100 reference phrases (`paraphrase-multilingual-MiniLM-L`
   as a candidate filter.
2. Layer 2 — binary NLI classification (`MoritzLaurer/multilingual-MiniLMv2-L6-mnli-xnli`), hypothesis picked in the caption's detected language.
3. Layer 3 — LLM (Ollama/Groq/Cerebras) performs event typing (16-label taxonomy), price-range extraction, and critically *gates* event creation based on `is_public_invitation/`

Report the false-positive reduction this gating produced, per the 100-post eval. **]**

## 3.7 Human Curation as the Trust Signal

**[TODO:** This is a genuine methodological pivot worth a dedicated section: the original automated account-scoring classifier (`old/1_harvest_account_classifier.py`) selected accounts where well under 10% turned out to be culturally valuable on manual review. `manualDataCuratedAt` (set only by `load_manual_account_categorization.py`) replaced the scoring-based selection as the operative "worth keeping" signal. Discuss this as a finding in itself: automated relevance scoring under-performed simple human judgment for this domain, and why (cite whatever intuition emerged — commercial/political accounts scoring artificially high, genuinely small cultural accounts scoring low on network centrality despite being sociologically important). ]

## 3.8 Geospatial Enrichment

**[TODO:** Nominatim geocoding, rate-limited (1.1s/req); hierarchy `Location`→`Arrondissement`→`City`→`Country`; three-tier confidence strategy (exact / city-combined / city-hint-only) and — **report honestly** — the empirical finding that low-confidence geocodes need explicit exclusion rather than fallback display, since a fallback to a default city hint can silently produce plausible-looking but wrong coordinates (found during dashboard validation, §3.11). ]

## 3.9 Ranking / Relevance Scoring

**[TODO:** Document the discovery-website ranking formula as a methodological artifact in its own right: Relevance $= 100 \cdot (0.30Q + 0.22A + 0.18B + 0.20T + 0.10C) \cdot$ penalty, where $Q$ is extraction quality/confidence, $A$ is network centrality (percentile-blended PageRank/betweenness/ followers/participation), $B$ is social resonance (hotness/post count), $T$ is temporal proximity, $C$ is session-based personalization with no login. Note the explicit removal of a nationality/ "Colombian-ness" bonus once the project's scope broadened — a methodological choice worth justifying (avoiding a nationality bias baked into a supposedly neutral relevance score). ]

## 3.10 Ethical Considerations

**[TODO:** Public-data-only scope; Instagram ToS; no scraping of private accounts; anonymization/aggregation choices for anyone appearing in captions/comments who is not an organizational account; data retention and the decision to eventually delete non-curated accounts (`cleanup_legacy_accounts.py`). ]

## 3.11 Limitations of Method

**[TODO:** Be concrete and honest — this section should draw on real bugs found during the dashboard validation, not hypothetical caveats: (1) geocoding fallback producing false-precision coordinates when a location could not be reliably resolved; (2) LLM date extraction misreading year/date from ambiguous captions (e.g. a caption stating only a month, defaulting to the wrong year); (3) Instagram-only sourcing missing organizations that primarily use other channels; (4) accounts outside the intended geographic scope (e.g. France-branded institutions physically located in Colombia) occasionally entering the graph via automatic relationship discovery before being caught by curation. **]**

# Chapter 4

# System Implementation: The Hub Cultural Platform

## 4.1 Pipeline Architecture Overview

**[TODO:** Reproduce the phase diagram; emphasize idempotency and incrementality as design principles running through every script. **]**

## 4.2 Extraction and Ingestion

**[TODO:** `1_harvest_ig_profiles.py` / `1_harvest_ig_posts.py` / `extract_profiles.py` → `2_build_graph.py`. Provenance tracking (`firstSeenAt`/`lastUpdatedAt`). **]**

## 4.3 Network Analysis Engine

**[TODO:** `3_analyze_network.py` internals; why it runs entirely offline from exported CSVs (portability, reproducibility benefit worth noting for a thesis — the analysis can be re-run without live database access). **]**

## 4.4 The Discovery Website

**[TODO:** Describe the client-facing artifact built from `5_export_dashboard_data.py` → static site (`site/`): dynamic category menu (a deliberate choice — no fixed editorial category list, so the menu reflects whatever the upstream human curation actually prioritizes), bilingual ES/FR interface, no-login session personalization via `localStorage`, similarity search via embeddings. Frame this as the thesis's applied contribution: turning the graph into something a member of the community could actually use, not just an analytical artifact for the researcher. **]**

## 4.5 Decision Log as Methodological Transparency

**[TODO:** Note the project's practice of a running, dated decision log (`docs/decisions.md`, DD-001 through DD-044+) as a form of methodological transparency — cite it as supporting material and summarize its most consequential entries (the curation pivot, the GDS→igraph pivot, the event-pipeline rewrite, the geocoding confidence gating fix). **]**

# Chapter 5

# Results

## 5.1 Network Structure

**[TODO:** Report current numbers precisely — re-pull from the latest `data_processed/*.csv` before writing, do not reuse V1 numbers unchanged (V1 reference point: ∼7,200 nodes seeded from a single account, 61 communities via Leiden, high modularity). State current account/event/location counts as of the most recent run (cf. latest export: 170 events, 200 accounts, at time of dashboard validation — confirm this is post-cleanup and reflects the broadened non-Colombia-only scope before citing). **]**

## 5.2 Bridge Actors and Community Structure

**[TODO:** The V1 sociological finding — commercial/institutional accounts show high E-I index (structural bridges) while individual accounts cluster locally — needs re-validation against the current graph (broadened scope, curation changes since V1). Report whether it still holds. **]**

## 5.3 Thematic and Event Landscape

**[TODO:** Distribution of events by category (musical, visual, institutional, festival, academic, etc.); how this compares to the four originally-targeted thematic agendas (literature, photography, cinema, theatre); note the empirical finding that `gastronomía`/institutional content over-represents relative to the curated thematic priorities, and how the dashboard's category menu and curation approach responded to that. **]**

## 5.4 Geographic Distribution

**[TODO:** Coverage: total events vs. events with a resolvable `Location` node vs. events with real lat/lon (report the actual, current, post-fix percentage — do not reuse the pre-fix 85.5% figure without re-checking after the geocoding-confidence exclusion fix described in §3.11). **]**

## 5.5 Data Quality Findings

**[TODO:** Treat this as a genuine result, not just a limitations footnote: quantify how often the geocoding fallback produced false-precision coordinates before the fix, how often LLM date extraction required manual correction, what fraction of institutional- category events were miscategorized on manual sample review (cf. the 20-event institutional sample: ∼35% either miscategorized or pipeline-gate leaks). This kind of transparent error analysis is strong thesis material. **]**

# Chapter 6

# Discussion

## 6.1 Interpreting Network Structure

**[TODO:** What does the bridge-actor pattern (if it holds) say sociologically about how this diaspora organizes culturally — mediated through a small number of institutional hubs rather than a dense peer network? **]**

## 6.2 From a Diaspora Network to a Cultural-Relevance Network

The project's scope changed twice. It began as a network of Colombian diaspora accounts, seeded from a single institutional account, the Colombian consulate in Paris. As the network grew through automatic relationship discovery (`RELATED_TO`), it broadened to Latin American diaspora accounts more generally. Both of these were still nationality-based criteria: an account or event qualified because of *who* was behind it.

The second, more consequential change replaced that criterion with a purely cultural one: an account or event qualifies if it is culturally relevant, with explicit priority given to photography, cinema, literature, theatre, and locally-organized community events, regardless of the nationality or cultural origin of the organizers behind it. This is a stricter methodological change than it might first appear, because it also required removing a nationality bonus that had previously been present in the discovery website's relevance-ranking formula (§3.9) — an explicit multiplicative bonus applied to events tagged with a Colombian cultural identity. Retaining that bonus after the scope change would have produced a ranking signal that was formally neutral (it ranks *cultural* events) while materially still favoring one nationality over every other one present in the same graph; removing it was necessary for the ranking formula to be honest about what it actually measures.
**[TODO:** Once network composition is re-pulled for Chapter 5, quantify here: what share of curated accounts/events are Colombian, other Latin American, French, or other, be-

fore vs. after the scope change; and revisit whether the four thematic pillars (literature, photography, cinema, theatre) still describe the bulk of curated content or whether the fifth, community-events category has grown large enough to warrant its own treatment in the results and in the discovery website's category menu. ]

What this change gained, beyond ranking-formula honesty, is a criterion that is easier to defend and to operationalize consistently: *is this cultural, and does it fall within one of these priority domains* is a judgment that can be applied uniformly to any account the discovery process surfaces. *Is this part of the Latin American diaspora*, by contrast, required an ad hoc call for every borderline case — a French venue that regularly hosts Colombian artists, a Latin American cultural institute whose visitors are mostly French — that the pipeline had no principled way to resolve consistently. What it cost is some of the project's original sociological specificity: the initial research question was, in part, about how *one* diaspora organizes culturally in a host country, and that question is necessarily diluted once the object of study is cultural activity in general. This thesis handles that trade-off by keeping the diaspora-specific reading as one lens applied to a subset of the network (wherever `culturalIdentity` data supports it), rather than as the network's defining boundary.

## 6.3 On Data Quality and the Limits of Automation

**[TODO:** Tie together §3.7 and §3.11: this project's most consistent lesson is that fully automated signals (account-relevance scoring, geocoding fallbacks, LLM date extraction) degrade silently and look confident even when wrong, while human curation — despite being unscalable — was the more reliable trust signal throughout. Discuss what this implies for similar future projects. **]**

## 6.4 Practical Implications: The Website as Intervention

**[TODO:** Beyond analysis, the discovery website is itself a sociological intervention — it could actually change how the diaspora finds/attends events. Discuss this as a design-research contribution, distinct from the analytical contribution. **]**

## 6.5 Methodological Reflections

**[TODO:** What would be done differently starting over — e.g. building the geocoding-confidence gate and human curation trust signal in from day one rather than arriving at them through iteration. ]

# Chapter 7

# Conclusion

## 7.1 Summary of Contributions

**[TODO:** Restate the three contributions from §1.4 concretely against what was actually delivered. **]**

## 7.2 Answering the Research Question

**[TODO:** Direct answer, one paragraph. **]**

## 7.3 Limitations

**[TODO:** Consolidate — do not just repeat §3.11 verbatim; frame as boundaries on how far the findings generalize. **]**

## 7.4 Future Work

**[TODO:** From the project's own V2 planning notes: a new NLP account classifier with explicit geography/cultural-relevance dimensions and anti-embeddings to penalize commercial/political content; expanded seed set from Latin American consulates in Paris; revised harvest cycle prioritizing cheap profile scraping before expensive post scraping; remaining dashboard work (clickable arrondissement/commune map, dedicated Explorer/Map/Organizer-profile pages, per-event bilingual content generated at extraction time, generic category image set). **]**

## 7.5 Closing Reflection

**[TODO:** Personal note tying back to §1.3's fourth objective — whether/how the project succeeded by its own most human metric (meeting someone through it), separate from the technical metrics. **]**

# Appendix A

# Graph Schema Reference

**[TODO:** Full node/relationship table, adapted from `docs/data_flow.md` and `CLAUDE.md`.
**]**

# Appendix B

# Selected Decision Log Entries

**[TODO:** Reproduce/summarize the most thesis-relevant entries from `docs/decisions.md` (curation pivot, GDS→igraph pivot, event-pipeline rewrite & 100-post eval, geocoding confidence gate). **]**

# Appendix C

# Discovery Website — Screenshots

**[TODO:** Home screen, filter bar, event detail panel — capture once the site is stable post-fixes (see current punch list: upcoming- filter bug, geocoding exclusion, account-scope cleanup). **]**

# Appendix D

# Repository Structure

**[TODO:** Reproduce the script naming convention and phase breakdown from `CLAUDE.md`'s Commands section as a reference table. **]**